

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Every datapoint in MILU originates from Indian competitive exams [Q9, Q11, Q20, Q24, Q93]. The deployment YAML and web search confirm structural absence of Bangladeshi content (GAP-1, GAP-6 [WEB-17, WEB-18]). Dataset analysis confirms this empirically: across 215 sampled examples, Bangladesh appears only as a wrong-answer distractor in two questions about Indian Pharmacopoeia [DATASET-D21, D25], and no Bangladeshi institution, ecological system, or trans-border policy appears anywhere in the sample (CRITICAL Concern 1). Bengali content is West Bengal exam-derived, with the entire validation Bengali sample being machine-translated from English (DATASET-D3, all 21 examples). Telugu content shows no Telangana/AP agro-ecological specificity (MAJOR Concern 6). For a HIGH-priority dimension targeting Bangladeshi farming context, Mymensingh dialect terminology, haor wetlands, and trans-border hydrology, the content is essentially zero — a fundamental misalignment.

ID	Evidence
Q9	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q11	"We create this benchmark by collecting questions from over 1500 competitive exams from India." (p.2)
Q20	"These questions were sourced following an approach similar to AGIEVAL (Zhong et al., 2023), collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams, among others." (p.3)
Q24	"Regional state exams are particularly valuable as they cover various state-level topics and emphasize the official language of each state." (p.3)
Q93	"We collected our questions from over 40 exam types ranging from various National and state level civil service examinations to examinations conducted by various government and private organizations." (p.17)
D3	**is_translated:** True — All 21 Bengali validation examples are machine-translated from English
D21	ଦ୍ରବିଡ଼େଶର 2019 ରେ ଭାରତୀୟ ଫାର୍ମାକୋପୋଇଆର ପ୍ରଥମେ ଜଣା ଦେଇଥିବା ଦେଶ କେଉଁ?... ଆଫଗାନିସ୍ତାନ; option2: ବାଙ୍ଗଳାଦେଶ — Which country first reported Indian Pharmacopoeia (Afghanistan) — Bangladesh appears as wrong answer option
D25	డిసెంబర్ 2019లో భారతీయ ఫార్మాకోపీయా ను గుర్తించిన మొదటి దేశం ఏదీ?... ఆఫ్ఘనిస్తాన్; option2: బంగ్లాదేశ్ — Same Indian Pharmacopoeia question in Telugu — Bangladesh again used only as distractor
WEB-17	AI4Bharat/MILU GitHub: no Bangladeshi annotator participation (https://github.com/AI4Bharat/MILU)
WEB-18	NAACL paper: annotation by AI4Bharat volunteers and Indian exam-portal experts only (https://aclanthology.org/2025.naacl-long.507/)

Strengths

- MILU does include English questions specifically chosen to address South Asian culture-specific content [Q26], offering some pan-South-Asian general knowledge value
- West Bengal-derived Bengali content shares deltaic ecology context with Mymensingh and may carry partial transferable agronomic relevance for the West Bengal cross-reference zone (though depth is unconfirmed)

ID	Evidence
Q26	"English questions are also included as these often address Indian culture-specific content, which is notably missing from existing popular benchmarks." (p.3)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires Bangladeshi agro-ecological knowledge (haor, beel, char-land, boro/aman cycles), Mymensingh dialect agricultural terminology, and cross-border GBM hydrology. None of this is present in the sampled MILU content (CRITICAL Concern 1) and the YAML confirms structural absence ([WEB-17] AI4Bharat/MILU GitHub).

IC-2: Cultural alignment is poor: MILU's Bengali corpus reflects West Bengal exam register conventions, not Bangladeshi register. The expert elicitation (A2) noted 'potentially slight implicit Indian Bengali stylistic bias' and confirmed dialect-specific Bangladeshi agricultural terminology is absent. Dataset analysis: 100% of sampled Bengali validation items are machine-translated [DATASET-D3], compounding register mismatch.

IC-3: Many inputs require Indian-specific knowledge that does not transfer: Indian constitutional procedure [DATASET-D5, D28], Indian state assembly elections [DATASET-D4], Indian medieval/colonial history [DATASET-D19, D38], Indian corporate brands [DATASET-D14, D33]. These contribute construct-irrelevant variance for an environmental-science evaluation.

IC-4: INSUFFICIENT DOCUMENTATION — no Bangladeshi annotators participated in MILU construction (GAP-6 resolution: 'CONFIRMED ABSENT' [WEB-17, WEB-18]). For the deployment, Bangladeshi agronomist review of any MILU-relevant subset would be required to flag culturally sensitive instances.

IC-5: Documented content issues: (a) zero Bangladesh-specific datapoints, (b) Bengali content sourced from West Bengal exams [Q24], (c) ~25% machine-translated overall [Q43] with the validation Bengali split observed at 100% translated [DATASET-D3], (d) Bangladesh appears only as distractor [DATASET-D21, D25]. These cumulatively violate content validity for the target deployment.

ID	Evidence
Q9	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q11	"We create this benchmark by collecting questions from over 1500 competitive exams from India." (p.2)
Q24	"Regional state exams are particularly valuable as they cover various state-level topics and emphasize the official language of each state." (p.3)
Q43	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
D3	**is_translated:** True — All 21 Bengali validation examples are machine-translated from English
D4	ফেব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিানসভা নির্বাচনে, আম আদমি পার্টি (AAP)... পাঞ্জাব — Indian state assembly election — Indian polity, no Bangladesh relevance
D5	নির্বাচন কমিশনারকে অপসারণ করা যত্নে পার... প্রধান নির্বাচন কমিশনারের সুপারিশে রাষ্ট্রপতি — Removal of Indian Election Commissioner — Indian constitutional law only
D14	নিম্নলিখিত পদ্ধতি দ্রুত কৃষি যন্ত্রপাতি উৎপাদন করে... কেরল কোম্পানি — Which Indian company produces agricultural equipment rapidly (Kirlskar) — India-specific corporate knowledge
D19	বাহমানি সাম্রাজ্যের প্রথম রাজধানী কী ছিল?... গুলবার্গা — First capital of Bahmani Kingdom — medieval Indian history, no agriculture/environment relevance
D21	তথ্যেবর 2019 ৱে ভারতীয় ঔষধিাদি ঔষধিাদি ঔষধিাদি... আফগানিস্তান; option2: বাংলাদেশ — Which country first reported Indian Pharmacopoeia (Afghanistan) — Bangladesh appears as wrong answer option
D25	ডিসেম্বর 2019-এ ভারতীয় ঔষধিাদি ঔষধিাদি ঔষধিাদি... অফিসিনালিস; option2: বাংলাদেশ — Same Indian Pharmacopoeia question in Telugu — Bangladesh again used only as distractor

D28	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया?... 44वें संविधान संशोधन अधिनियम द्वारा — Indian constitutional amendment for 'armed rebellion' clause — Indian law and governance
D33	T - Hub ଏକ ଡିଜିଟାଲ ରାଜ୍ୟ ସରକାରର ପ୍ରୟୋଗ... ଉତ୍ପାଦନା ନୁହେଁ ବରଂ ଉତ୍ପାଦନ ଦେବାକୁ ଏକ ପ୍ରୟୋଗ ଉପରେ କେନ୍ଦ୍ର — T-Hub Telangana state entrepreneurship tech incubator — Telangana-specific but non-agricultural
D38	प्रांतांमध्ये द्वैधशासन प्रणाली कोणत्या कायद्याने स्थापन केली?... भारत सरकार अधिनियम 1919 — Government of India Act 1919 establishing dyarchy — Indian colonial law
WEB-17	AI4Bharat/MILU GitHub: no Bangladeshi annotator participation (https://github.com/AI4Bharat/MILU)
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Information Gaps

- Whether the test split (vs validation) contains a higher proportion of original (non-translated) Bengali questions (DATASET limitation 1)
- Quantitative subject-level question counts for Agriculture/Environmental Science in Bengali and Telugu test splits (GAP-4 not resolved [WEB-18])

Requires Expert Verification

- A Bangladeshi linguist's assessment of register/terminology drift in MILU's machine-translated Bengali items

Remediation

Gap 1: Zero Bangladeshi datapoints; Bengali corpus is West Bengal-derived; sampled Bengali validation is 100% machine-translated

Recommendation: Source new Bengali items from Bangladeshi institutional question banks (BAU, BARI, BRRI, BWDB) authored in Bangladeshi register; integrate BEnQA [WEB-6] as a complementary Bangladeshi Bengali baseline; explicitly flag and quarantine machine-translated items in any deployment screening report.

ID	Evidence
WEB-6	BEnQA exists as a separate Bangladeshi Bengali science benchmark, but is not integrated into MILU (https://arxiv.org/abs/2403.10900)